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Measurements of Plasma Properties Using Fast Sweep Langmuir Probes in Unmagnetized Weakly Ionized Plasmas

O. Troll[2], **L. Conde**[*1], **E. Criado**[1], **J. M. Donoso**[1], and **G. Herdrich**[2]

[1] Departamento de Física Aplicada, E.T.S.I. Aeronáuticos. Univ. Politécnica de Madrid, 28040 Madrid, Spain
[2] Institut für Raumfahrtsysteme (IRS). Universität Stuttgart. Pfaffenwaldring 31, D-70550. Stuttgart. Germany.

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The performances of a simple circuit for fast sweep measurements using collecting and emissive Langmuir probes are evaluated. The probes are biased by means of a time dependent ramp voltage signal with a variable pulse frequency and the current voltage curves are measured along the increasing flange of this sawtooth voltage. The response of this fast probe polarization circuit was verified under actual experimental conditions by measuring the properties of a stationary Maxwellian plasma produced by means of a glow discharge. The results are independent of the experimental conditions and essentially rely on the discharge properties for polarization pulse repetition rates below a threshold. This upper bound lies below the ion plasma frequency and is related with the faster time scale involved in the sawtooth signal probe bias voltage. The motion of ions would not follow the rapid change of the electric field around the probe associated to the short decreasing edge of the sawtooth polarization voltage and, therefore, the probe perturbs the local electric field. We conclude that these time scales should be considered for the interpretation of these measurements in addition to the electron and ion plasma frequencies in fast sweep Langmuir probe techniques. We conclude that these time scales should be considered for the interpretation of these measurements in addition to the electron and ion plasma frequencies in fast Langmuir probe techniques.

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1 Introduction

The diagnose of low temperature plasmas using collecting and emissive Langmuir probes (LP) is a widespread technique [1, 2]. The collecting LP are electrodes, with a well defined shape (planar, cylindrical or spherical) which are immersed into the plasma. In accordance to its bias potential V_p with respect to the local plasma potential V_{sp}, its exposed metallic surface repels the electrons or ions and drains an electric current from the plasma $I_p(V_p)$. The electron temperature $k_B T_e$, density n_e and local plasma potential V_p are calculated from these current voltage (IV) characteristic curves [1–4].

The emissive LP is essentially employed to measure the local plasma potential and is made up of a thin metallic wire also exposed to the plasma. [1–3, 5]. This wire is heated up to the electron thermoionic emission using a DC current and is electrically biased as the collecting probes. Therefore, the electrons are repelled for $V_p < V_{sp}$ and attracted for $V_b > V_{sp}$ and this emitted thermoionic electron current appears as an effective ion current, added to the collected current by the wire. For large enough wire temperatures this emitted electron current produces a steeper transition in the IV curve. Then, this transition from electron emission to collection creates an abrupt jump in $I_p(V_p)$ that allows an accurate determination of the plasma potential V_{sp}, currently determined from the IV curves by means of either, the *floating point* or the *inflection point* methods [1, 3–5].

Both, collecting and emissive Langmuir probes provide local measurements and are useful to determine the space dependent properties of low density plasmas. They are employed to measure the space dependent properties of plasma plumes produced by ion thrusters or the plasma potential profile of double layers [6]. Their spatial resolutions is essentially limited by the plasma Debye length λ_D and physical size of the probe d and they are usually employed when $\lambda_D/d \ll 1$. However, their time response is poor and it is basically limited by bias voltage sweep rate used to obtain the IV curves.

* Corresponding author: E-mail: luis.conde@upm.es

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In consequence, different configurations have been proposed in order to improve the low time resolution of Langmuir probes, as fast sweep bias circuits for collecting single tip probes [7]. When the bias sweep rate increases over 100 Hz, the impedance coupling significantly increases the levels of noise that could be reduced by using double probe schemes. In addition to the plasma exposed tip an additional mirror dummy probe is used to compensate the impedances [8–10]. More recently, this mirror probe has been replaced by electronic circuits that overcome this impedance unbalance [11].

In this paper we evaluate the performances of a simple electronic circuit which permits fast sweep measurement of voltage current characteristic (IV) curves using either, collecting and emissive Langmuir probes. In order to determine the response under actual experimental conditions, the probes were used to diagnose a glow discharge plasma. This electric discharge provides a stable and uniform Maxwellian plasma column suitable for repetitive and systematic testing.

The probes are polarized by applying a time dependent voltage ramp with sweep frequencies up to 2 MHz and the IV curves are measured along the increasing voltage side. This permits to resolve the temporal evolution of the IV curves with sampling frequencies over the typical ion plasma frequencies f_{pi} of the glow discharge. As we shall see, the practical limit of fast sweep Langmuir probe technique falls below f_{pi} and is associated to the shortest time scale involved in the voltage ramp polarization of the Langmuir probes.

2 Experimental setup

The scheme of the experimental setup is shown in Fig. 1 together with the circuit for fast sweep bias of the Langmuir probes. The neutral gas pressure inside the vacuum chamber was held constant by means of a controlled leak of Argon using a Bronkhorst E7100 controller with a F-210C valve between 4×10^{-3} and 8×10^{-3}. The glow discharge plasma was produced by applying a high voltage between the walls of an stainless steel vacuum chamber (the cathode) and a metallic plate located inside the chamber volume (the anode). The discharge voltages were in the range $V_{ds} \simeq$ 400-500 V, for the discharge currents of I_{ds} = 10-50 mA while the ballast resistance R_B was between 1-2 KΩ.

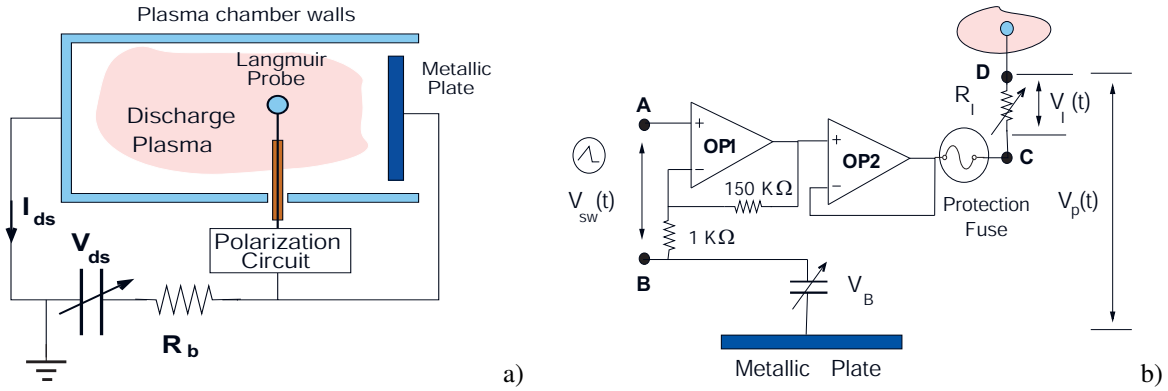


Fig. 1 The scheme of the experimental set up (**a**) and the details of the circuit employed for fast sweep polarization of Langmuir probes (**b**). (Color figure: www.cpp-journal.org).

As indicated in the scheme of Fig.1 **a**, the emissive and collecting Langmuir probes were placed inside the positive column of the glow discharge plasma. The collecting probe employed in these experiments was a stainless steel cylinder of 0.4 mm of radius and 22.4 mm in length. The emissive probes were made of a 0.08 mm diameter tungsten wire connected to two leads inside a ceramic tube and heated up to the strong thermoionic emission by a 0.5-2.0 A current using an DC stabilized power supply. Finally, the electronic temperature $k_B T_e$, density n_e, and the plasma potential V_{sp} are later calculated from the digitized IV curves using standard methods [1, 2].

Both, collecting and emissive probes are biased using the circuit of Fig.1 **b** which uses two Burr Brown 3583AM operational amplifiers. The amplifier $OP1$ is configured to add the DC polarization potential V_B to the amplified time dependent sawtooth voltage $V_{sw}(t)$ from the signal generator (Kenwood FGE 1021). The second amplifier $OP2$ is a voltage follower which applies to the probe the bias potential $V_p(t) = V_B + V_I(t) + G \times V_{sw}(t)$.

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The measured gain of the first amplifier circuit was $G = 13.5$ and this value of G was found constant for sweep frequencies f_{sw} between 100 Hz up to 2 MHz. In addition, the observed distortion introduced by the circuit of Fig. 1 **b** in the amplified signal $V_p(t)$ was found negligible when was compared with $V_{sw}(t)$ within this frequency range.

The scheme with the time dependent probe bias voltage is shown in Fig. 2 **a**. In order to reduce the time lag τ between two successive measurements of the IV curves, the bias voltage $V_{sw}(t)$ was adjusted in six fixed steps (S1-S6). This covers from a perfect triangular symmetry up to a ramp sawtooth signal with the minimum descending flange allowed by the signal generator. For a fixed signal period T_{sw} the time elapsed along the rising and falling edges of a symmetric triangular (S1) signal $V_{sw}(t)$ is equal $t_{S1} = \tau_{S1} = 1/(2 \times f_{sw})$. For the longest ramp signal (S6) the time for increasing voltages grows up to $t_{S6} = 1.78 \times t_{S1}$ while the time of the decreasing flange becomes $\tau_{S6} = T_{sw} - t_{S6}$.

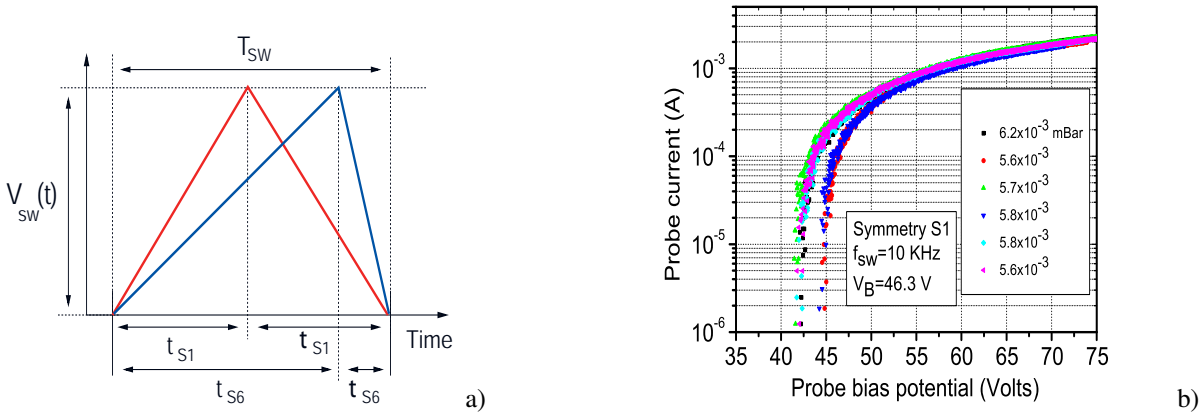


Fig. 2 The scheme with the polarization signal of the Langmuir probes (**a**) and the typical results for the triangular probe symmetry (S1) for the fixed sweep frequency of 10 KHz (**b**). (Color figure: www.cpp-journal.org)

The probe current $I_p(t) = V_I(t)/R_I$ is measured along the increasing bias voltage flange of Fig. 2 **a** and calculated from the voltage drop $V_I(t)$ across the resistor $R_I = 5.039$ KΩ. All electric equipments in the bias circuit of Fig. 1 **b** were made electrically independent using insulation transformers and the points A and C in Fig. 1 **b** are at the same electric potential. This permits to fed both time dependent voltage signals $V_{sw}(t)$ and $V_I(t)$ into a Yokogawa DL9140 digital scope where are stored for their subsequent analysis. Therefore, the time fluctuations in the probe current $I_p(t)$ could be resolved up to the maximum bandwidth (1 GHz) of this digital scope.

3 Experimental results and discussion

The typical IV curves for collecting Langmuir probes obtained using the fast sweep polarization circuit of Fig.1 **a** are shown in Fig. 2 **b** and Fig. 3. For sweep bias frequencies $f_{sw} < 10$ KHz and similar values of the discharge parameters (V_{ds} and I_{ds}) the results of the measurements are independent of both, the neutral gas pressure and the symmetry (S1-S6) of the driving signal $V_{sw}(t)$.

The only differences observed in the IV curves of collecting LP rely in the value of the plasma potential which slightly changes during the experiments as in Fig. 2 **b**. In addition, the levels of noise observed in the measurements of the probe currents are low, as evidence Figs. 2 **b** and Fig. 3. These experimental results are in agreement with the properties of a glow discharge plasma. The involved electron temperatures were found between $k_B T_e \simeq 0.95$-1.3 eV and the plasma densities were within the range of $n_e \simeq 3.0 - 6.5 \times 10^8$ cm^{-3}.

For a reference electron density of 4.8×10^8 cm^{-3} the electron and ion plasma frequencies are about $f_{pe} \simeq 0.2$ GHz and $f_{pi} \simeq 730$ KHz in these experiments. Therefore, the probe bias sweep frequencies $f_{sw} < 10$ KHz in Figs. 2 **b** and Fig. 3 lie below the characteristic ion plasma frequencies which are also over the maximum circuit sweep frequency of 2 MHz.

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On the contrary, for sweep frequencies $f_{sw} \geq 10$ KHz the IV curves of collecting LP of Fig. 3 appear as deformed when they are compared with those of a Maxwellian plasma of Fig. 2 **b**. We interpret these anomalies as originated by the slow response of ions during the time scale of the falling edge of the applied bias voltage $V_{sw}(t)$. The only difference in the experimental conditions of Figs. 2 **b** and 3 **a** is the shorter falling edge of the ramp sweep signal $V_{sw}(t)$. As the scheme of Fig. 2 **a** shows, for the triangular symmetry (S1) and $f_{sw} = 10$ KHz the time for the increasing polarization voltages of the probe is $t_{S1} = 0.05$ ms which grows up to $t_{S6} = 1.78 \times t_{S1} = 0.088$ ms for the sawtooth ramp symmetry (S6). For a constant wave period of $T_{sw} = 0.1$ ms the duration of the falling edge of $V_{sw}(t)$ reduces from $\tau_{S1} = t_{S1} = 0.05$ ms down to $\tau_{S6} = 0.012$ ms.

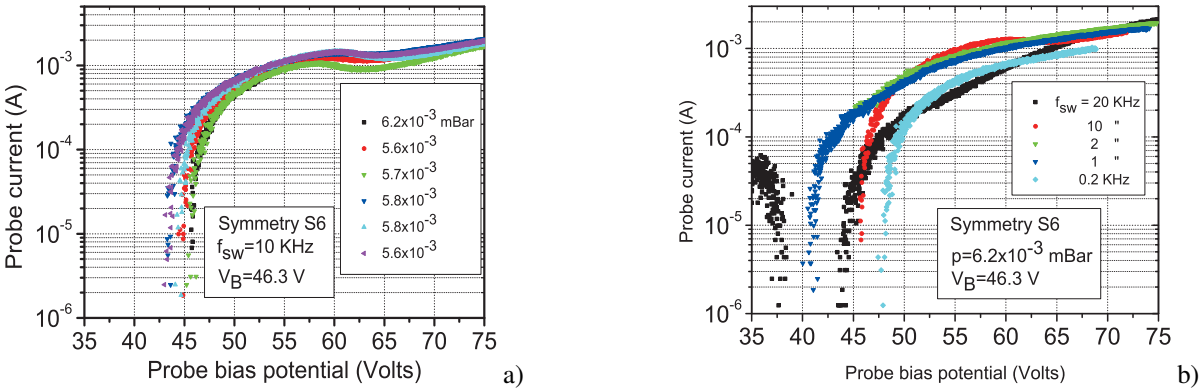


Fig. 3 Measurements of the IV curves of collecting Langmuir probes biased by a voltage ramp signal of symmetry (S6); with sweep frequency of 10 KHz (Fig. **a**) and for different sweep frequencies (Fig. **b**). (Color figure: www.cpp-journal.org).

In this glow discharge plasma, the shortest time scale for ion motion is given by $\tau_{pi} = 1/f_{pi} \simeq 0.0014$ ms. For the rising edge of $V_{sw}(t)$ we have, $\tau_{S1}/\tau_{pi} \simeq 36$ and $\tau_{S6}/\tau_{pi} = 8.5$. Therefore, despite the plasma ion frequency f_{pe} is higher than f_{sw}, the short time response associated with a voltage ramp probe polarization τ_{S6} lies close to the typical time scale τ_{pi} for ion motion.

This effect is evidenced in Fig. 3 **b** for the symmetry S6, where for low sweep frequencies (0.2-2 KHz) the IV curves correspond to a Maxwellian plasma while becomes deformed for $f_{sw} \geq 10$ KHz. The IV curves in Fig. 2 **b** deforms for decreasing τ_{S6} times as f_{sw} grows while the plasma parameters remained unchanged. These results are independent of the neutral gas pressures as Fig. 3 **a** shows. The experimental results of Fig. 3 suggest that the ions cannot return back to their initial state during τ_{S6} and therefore, the local plasma becomes perturbed along this fast time scale.

The typical IV curves for emissive Langmuir probes in Fig. 4 are similar to those found by other authors [3,5]. The increment of the DC power increases the heating voltage V_H, raising the temperature of the wire and this produces the thermoionic electron emission that appears as an effective ion current for bias voltages V_p below the plasma potential. The magnitude of this electron emitted current for $V_p < V_{sp}$ increases with V_H and the floating potential V_F of the IV curve approaches V_{sp}.

However, for positive bias voltages $V_p > V_{sp}$ the electron saturation current increases with the temperature of the wire. This response has been frequently observed in the operation of emissive Langmuir probes and is not yet fully understood [5]. The possible suggested causes are, the charge space effects, the increasing rejection of contaminants by the heated probe or the physical growth of the effective electron collection surface by the thermal expansion of the tungsten wire [5]. Nevertheless, there is not an accepted explanation for this increment in the electron saturation current associated to high wire temperatures at the present time.

For $V_H < 3.0$ V the IV curves of emissive probes in Fig. 4 present higher levels of noise than those found in collecting probes, which are independent of the magnitude of the heating voltages. These high frequency fluctuations are evidenced because of the large bandwidth of the probe bias circuit of Fig. 1 **b** and result averaged when low frequency data acquisition systems are employed. Unfortunately, we are not in position to explain the origin of these oscillations in the probe current, which are not present in the IV curves of the collecting probes, and therefore are not introduced by the polarization circuit of Fig. 1 **b**.

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The results of Fig. 4 correspond to two successive operations of the emissive probe under equivalent discharge conditions and then evidence the physical degradation of tungsten wire exposed to the plasma. The large amplitude fluctuations in the probe current for higher wire temperatures ($V_H > 3.0$ V) observed for $V_p > V_{sp}$ in Fig. 4 **b** are not present in the first measurements of Fig. 4 **a**. These results suggest that the wire emits to the plasma positively charged tungsten particles in addition of the electron collected that produces the physical degradation of the exposed wire surfaces.

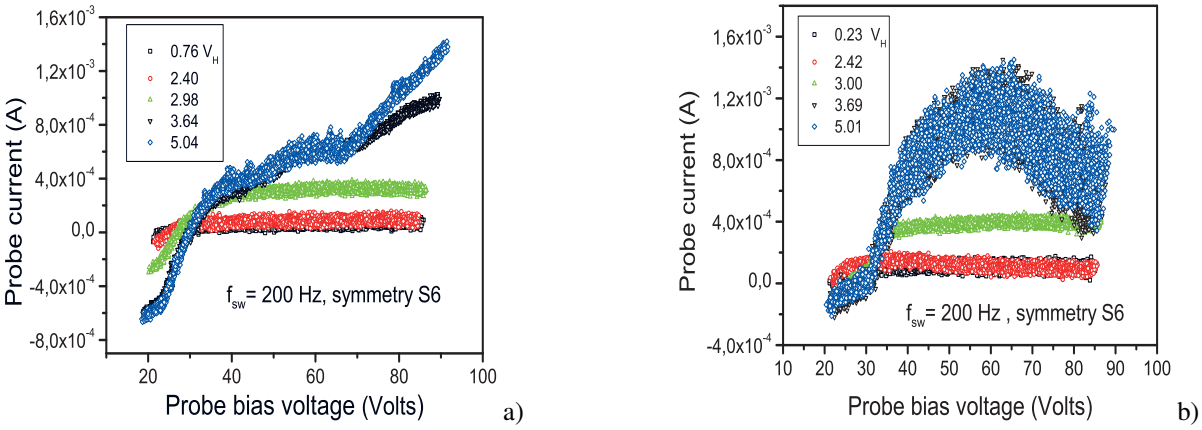


Fig. 4 The IV curves for an emissive Langmuir probe for different heating voltages V_H. The argon gas pressure was 6.6×10^{-3} mBar, $V_{ds} = 445$ V and $I_{ds} = 20$ mA. (Color figure: www.cpp-journal.org).

In conclusion, the fast sweep bias circuit of Fig. 1 **b** provides reliable measurements of the IV curves using either, collective and emissive Langmuir probes. The deformations observed for bias sweep frequencies f_{sw} over an onset value under the same plasma conditions in the curves of Fig. 3 suggest that the collecting probe perturbs the surrounding plasma. These experimental results indicate that this effective upper bound for the maximum frequency sweep, is not caused by the limitations of the circuit of Fig. 1 **b**. The slow response of ions to fast bias voltage ramps influences the development of the plasma sheath and therefore their orbital motion is hindered by the fast change of the electric field around the Langmuir probe. This inertial effect of heavy particles would be enhanced by the collisions of ions with neutrals which confine the massive ions. In consequence, in addition to the electron and ion plasma frequency, the time scales involved in fast sweep Langmuir probe techniques should be considered for the interpretation of these measurements.

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